/* 2) Write a C program that contains a string (char pointer) with a value "Helloworld". The program should AND or XOR each character in this string with 127 and display the result. */

```
/* ===== 1) XOR ===== */

#include <stdio.h>
#include <string.h>

int main()
{
   char String[] = "Helloworld";
   char cipher[100];
   char plain[100];
   int len = strlen(String);

   for (int i = 0; i < len; i++)
   {
      cipher[i] = String[i] ^ 127;
   }
   cipher[len] = '\0';
   printf("Cipher text: %s\n", cipher);

   for (int i = 0; i < len; i++)
   {
      plain[i] = cipher[i] ^ 127;
   }
   plain[len] = '\0';
   printf("Plain text: %s\n", plain);

   return 0;
}

/* Output:-
Cipher text: 7<non-printable characters>
Plain text: Helloworld
*/


/* ===== 2) AND ===== */

#include <stdio.h>
#include <string.h>

int main()
{
   char String[] = "Helloworld";
```

```
    char cipher[100];
    char plain[100];
    int len = strlen(String);

    for (int i = 0; i < len; i++)
    {
        cipher[i] = String[i] & 127;
    }
    cipher[len] = '\0';
    printf("Cipher text: %s\n", cipher);

    for (int i = 0; i < len; i++)
    {
        plain[i] = cipher[i] & 127;
    }
    plain[len] = '\0';
    printf("Plain text: %s\n", plain);

    return 0;
}

/* Output:-
Cipher text: Helloworld
Plain text: Helloworld
*/
```

4) Write a program to calculate SHA-512 value.

```
import java.math.BigInteger;
import java.security.MessageDigest;
import java.security.NoSuchAlgorithmException;
import java.util.*;

public class SHA512 {
    public static void main(String args[]) throws NoSuchAlgorithmException
    {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter input String(Plain text Message): ");
        String str = sc.next();
        String hashcode;
        MessageDigest md;
        try {
            md = MessageDigest.getInstance("SHA-512");
            byte[] messageDigest = md.digest(str.getBytes());
            BigInteger no = new BigInteger(1, messageDigest);
            hashcode = no.toString(16);
            while (hashcode.length() < 32) {
```

```
            hashcode = "0" + hashcode;
          }
       }
       catch (NoSuchAlgorithmException e) {
          throw new RuntimeException(e);
       }
       System.out.println("HashCode Generated by SHA-512 for: ");
       System.out.println("\n" + str + " is : " + hashcode);
       System.out.println("Message digest object info: ");
       System.out.println(" Algorithm = " + md.getAlgorithm());
       System.out.println(" Provider = " + md.getProvider());
       System.out.println(" ToString = " + md.toString());
    }
}
```

Output:-

```
Enter input String(Plain text Message):
hello
HashCode Generated by SHA-512 for:

hello is :
9b71d224bd62f3785d96d46ad3ea3d73319bfbc2890caadae2dff72519673ca72323c3d99ba5
c11d7c7acc6e14b8c5da0c4663475c2e5c3adef46f73bcdec043
Message digest object info:
 Algorithm = SHA-512
 Provider = SUN version 17
 ToString = SHA-512 Message Digest from SUN, <initialized>
```

10) Write a Java program to perform Encryption and Decryption using Substitution Cipher.

```
import java.util.Scanner;

public class SubstitutionCipher {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";

        System.out.print("Enter substitution key: ");
        String key = sc.nextLine().toUpperCase();

        System.out.print("Enter plain text: ");
        String plaintext = sc.nextLine().toUpperCase();

        String encrypted = "";
        for (int i = 0; i < plaintext.length(); i++) {
            char ch = plaintext.charAt(i);
```

```
            int index = alphabet.indexOf(ch);
            if (index != -1) {
                encrypted += key.charAt(index);
            } else {
                encrypted += ch;
            }
        }
        System.out.println("Encrypted text: " + encrypted);

        String decrypted = "";
        for (int i = 0; i < encrypted.length(); i++) {
            char ch = encrypted.charAt(i);
            int index = key.indexOf(ch);
            if (index != -1) {
                decrypted += alphabet.charAt(index);
            } else {
                decrypted += ch;
            }
        }
        System.out.println("Decrypted Text: " + decrypted);

        sc.close();
    }
}
```

Output:-

```
Enter substitution key: qwertyuiopasdfghjklzxcvbnm
Enter plain text: kits
Encrypted text: AOZL
Decrypted Text: KITS
```